Data Science - 22DS102006

2. Interacting with Web APIs and Databases

a. GitHub Issues to DataFrame

```
import requests
import pandas as pd

url = 'https://api.github.com/repos/pandas-dev/pandas/issues'
response = requests.get(url)
issues = response.json()
df_issues = pd.DataFrame(issues)
print(df_issues[['id', 'title', 'state', 'created_at']].head())
```

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Step by Step Explanation

- `requests.get(url)`: Sends a **GET request** to GitHub's API to retrieve the issues of the `pandas-dev/pandas` repository.
- `.json()`: Parses the JSON response into a Python object (usually a list of dictionaries).
- `pd.DataFrame(issues)`: Converts the list of issue dictionaries into a **DataFrame**.
- `print(...)`: Displays the first 5 issues, showing selected columns: `id`, `title`, `state`, and `created_at`.

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b. SQLite Database Operations

```
import sqlite3
import pandas as pd

# Connect to database
conn = sqlite3.connect("students.db")
cursor = conn.cursor()

# Create table
cursor.execute('''CREATE TABLE IF NOT EXISTS students (id INTEGER, name TEXT)''')

# Insert datacursor.execute("INSERT INTO students VALUES (1, 'John')")
conn.commit()
cursor.execute("INSERT INTO students VALUES (2, 'Joy')")
conn.commit()

# Read datadf = pd.read_sql_query("SELECT * FROM students", conn)
print(df)

# Update and Delete
cursor.execute("UPDATE students SET name = 'Alice' WHERE id = 1")
cursor.execute("DELETE FROM students WHERE id = 1")
conn.commit()
conn.close()
```

/*

Explanation

- `sqlite3.connect(...)`: Connects to a database file (`students.db`). If it doesn't exist, it creates one.
- `cursor.execute(...)`: Executes SQL queries.
- `CREATE TABLE IF NOT EXISTS`: Ensures the table is created only once.
- `INSERT INTO`: Adds a new student with `id = 1`, `name = John`.
- `pd.read_sql_query(...)`: Reads the table as a DataFrame for easy display and analysis.
- `UPDATE`: Changes the student's name to `Alice`.
- `DELETE`: Removes the student record from the table.
- `conn.commit()`: Saves changes to the database.
- `conn.close()`: Closes the connection.

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